

GenAI on GCP - DocuMind course: modules and lessons

Plan v5, 22 September 2026. 18 modules, 60 lessons: 14 Core modules with 47 lessons, 4 Advanced modules with 13 lessons. Each module carries a one-word name, then what the learner does in it.

Module	Lesson	Lesson title	Track
1 • Setup Run and understand DocuMind	1.1	Reproduce the local environment and read the master diagram	Core
	1.2	Prove a first success and diagnose a first failure	
2 • Deployment Use the cloud lab safely	2.1	Understand the project, identities and resource map	Core
	2.2	Review and start the prepared deployment	
	2.3	Check readiness, save progress and close the session	
3 • Ingestion Turn documents into searchable evidence	3.1	Understand source, tenant, page and chunk contracts	Core
	3.2	Parse documents and compare chunk boundaries	
	3.3	Create and validate compatible embeddings	
	3.4	Write, inspect and verify indexed records	
4 • Lifecycle Deliver, update and recover documents	4.1	Follow upload events, retries, dead-letter handling and the batch lane	Core
	4.2	Reindex a changed section and measure embedding reuse	
	4.3	Publish versions, retire documents and reject stale events	
	4.4	Restore documents and reconcile index differences	
5 • Retrieval Retrieve the right evidence	5.1	Apply query embeddings and authorized filters	Core
	5.2	Compare dense and hybrid retrieval	
	5.3	Rerank and inspect retrieved candidates	
	5.4	Test fallback without losing tenant or metadata filters	
6 • Generation Produce grounded answers and a usable interface	6.1	Pack evidence within the context budget	Core
	6.2	Generate structured answers, citations and refusals	
	6.3	Stream answers and handle failures	
	6.4	Complete the Streamlit upload-to-answer journey	
7 • Evaluation Measure quality and reject regressions	7.1	Build a useful evaluation dataset	Core
	7.2	Separate offline checks, live scoring and LLM judgment	
	7.3	Compare one controlled change against a baseline	
8 • Security Verify identity, tenant isolation and safety	8.1	Trace authenticated identity into tenant membership	Core
	8.2	Test valid access, denied access and cross-tenant requests	
	8.3	Exercise DLP, guardrails and audit behavior	
9 • Caching Cache without serving stale answers	9.1	Compare context caching and answer caching	Core
	9.2	Test cache scope, configuration changes and freshness	
	9.3	Measure latency, avoided calls and false cache hits	

Module	Lesson	Lesson title	Track
10 • Agents Build a tool-using agent	10.1	Understand tool contracts and the direct agent loop	Core
	10.2	Implement the main LangGraph workflow	
	10.3	Diagnose tool arguments, access failures and timeouts	
	10.4	Compare the LangChain and ADK adapters	
11 • Memory Persist and isolate conversation state	11.1	Distinguish agent state, conversation history and knowledge	Core
	11.2	Configure and inspect durable conversation storage	
	11.3	Verify restart recovery and session isolation	
12 • Protocols Connect external agents through MCP and A2A	12.1	Expose, discover and invoke MCP tools	Core
	12.2	Deploy MCP and verify authorized access	
	12.3	Trace the implemented A2A peer and its permissions	
13 • Operations Diagnose failures and control spending	13.1	Debug a wrong answer through the complete pipeline	Core
	13.2	Reconcile usage events, reports and alerts	
	13.3	Exercise model routing, budgets and shutdown controls	
14 • Release Release, recover and demonstrate the project	14.1	Build and deploy using keyless identity	Core
	14.2	Evaluate and gate the exact candidate revision	
	14.3	Promote, roll back and repair a controlled failure	
	14.4	Complete the independent capstone and operational handover	
15 • Graph Graph and managed retrieval	15.1	Build graph evidence from source documents	Advanced
	15.2	Compare Firestore and Spanner graph paths	
	15.3	Configure and query managed retrieval mirrors	
	15.4	Compare quality and test update/withdrawal freshness	
16 • Multimodal Multimodal evidence and media	16.1	Query a video clip and validate media citations	Advanced
	16.2	Exercise implemented Studio and voice features	
17 • Tuning Managed fine-tuning	17.1	Decide whether tuning is justified and prepare data	Advanced
	17.2	Validate sanitized datasets and run managed tuning	
	17.3	Compare the tuned candidate with an uncontaminated baseline	
18 • Serving Model gateway and self-hosted inference	18.1	Trace and authorize gateway routes	Advanced
	18.2	Serve a supplied or stock model using Ollama	
	18.3	Inspect the vLLM service and the GKE alternative	
	18.4	Compare actual backends and verify shutdown behavior	